Measuring sidewall angle optically with test structures

What the project is about

Silicon photonic chips are very sensitive to small fabrication errors. A few nanometers of error in waveguide width or thickness, or a degree or two of sidewall tilt, noticeably shifts how a device behaves. Foundries keep track of this by putting test structures on the wafer (ring resonators, Mach-Zehnder interferometers, plain waveguides) and measuring them optically after fabrication. Width and thickness can be pulled out of these measurements fairly well. Sidewall angle is much harder, and in practice people usually fall back on cutting the wafer and looking at it in an SEM, which is slow and destroys the sample.

This isn't just my impression. Sidewall angle varies across a wafer because of etch loading, one photonic-crystal cavity study traced lower than expected Q-factors to non-vertical sidewalls, and a 2023 compact-modeling paper found that its optical extraction fit was essentially flat across sidewall angles from 85° to 90°, meaning the measurements could not tell those angles apart.

The test structures in use today are picked by experience, not by any formal measure of how much they actually tell you. My idea was to treat this as an experiment-design problem: for a given set of test structures, work out how well width, thickness and sidewall angle could in principle be recovered, and then look for structures that improve that, especially for sidewall angle.

How I did it

For each candidate structure I solved for the optical mode of the waveguide cross-section with a full-vector finite-difference mode solver (EMpy). I then nudged the width, thickness and sidewall angle by small amounts (5 nm, 2 nm and 1°) and recorded how the effective index and group index changed. Those derivatives show how strongly each measurement responds to each fabrication parameter.

I combined the derivatives with an assumed measurement noise level for each readout (a ring gives an effective index and a group index, an MZI gives a phase) to build a Fisher information matrix. Inverting it gives the best precision any extraction could achieve for each parameter (the Cramér-Rao bound). Comparing that number across different sets of structures shows which structures actually add information instead of repeating what is already there.

The solver was checked against published values for the standard 500 x 220 nm silicon waveguide (effective index about 2.44), against an independent effective-index calculation I wrote myself, and against limiting cases (a rib with zero slab should reproduce a strip exactly, which it does). Structures covered: strip waveguides (TE and TM), slot waveguides (TE and TM), rib waveguides (TE) and a subwavelength grating (TE), each at five widths, giving 30 structures in total.

What I originally found

Rings alone don't do it

Two TE rings of different widths were badly conditioned for recovering all three parameters together, with condition numbers between 2.4e5 and 7.3e9. Sidewall angle sensitivity barely changes from one ring width to the next, so its effect looks almost the same as a mix of width and thickness changes. The choice of ring widths still mattered a lot: the condition number varied about 30,000x between the best and worst pair.

The condition number was the wrong yardstick

At first, adding TM rings or MZIs seemed to make things worse, because the condition number went up. Looking at the smallest eigenvalue of the Fisher matrix instead (which is what actually limits the worst-determined parameter) showed that every added structure helped, and the MZI helped the most. After this I used the smallest eigenvalue and the trace of the inverse as the main metrics.

Added to the best TE ring pair (550 + 600 nm)	Smallest eigenvalue	Condition number
Nothing (baseline)	6.33	2.4e5
Another TE ring (400 nm)	7.64	3.0e5
TM ring (400 nm)	6.96	3.6e5
MZI (600 nm, TE)	10.21	6.6e5

One catch with MZIs: at the same width and polarization as a ring, the MZI phase is just a scaled copy of the ring's effective-index measurement, so it only adds something when paired with a different width or polarization.

Different structures point in different directions, but that alone isn't enough

I turned each structure's derivatives into a direction in (width, thickness, angle) space and compared them (Figure 1). Structures of the same type at different widths are nearly parallel (mean angle between them about 4°), while different types are clearly separated (mean about 46°), and the structures cluster cleanly by type.

The surprise was that looking different didn't predict being useful. The angle between two structures' directions actually correlated with the wrong sign against the Fisher result (Spearman -0.37 against the smallest eigenvalue). What mattered was how much each structure on its own pointed along the sidewall-angle axis. Slot TE had the largest angle component of the early structures (-0.60 to -0.62), which is why it helped. Adding group-index information to the direction vectors didn't change this.

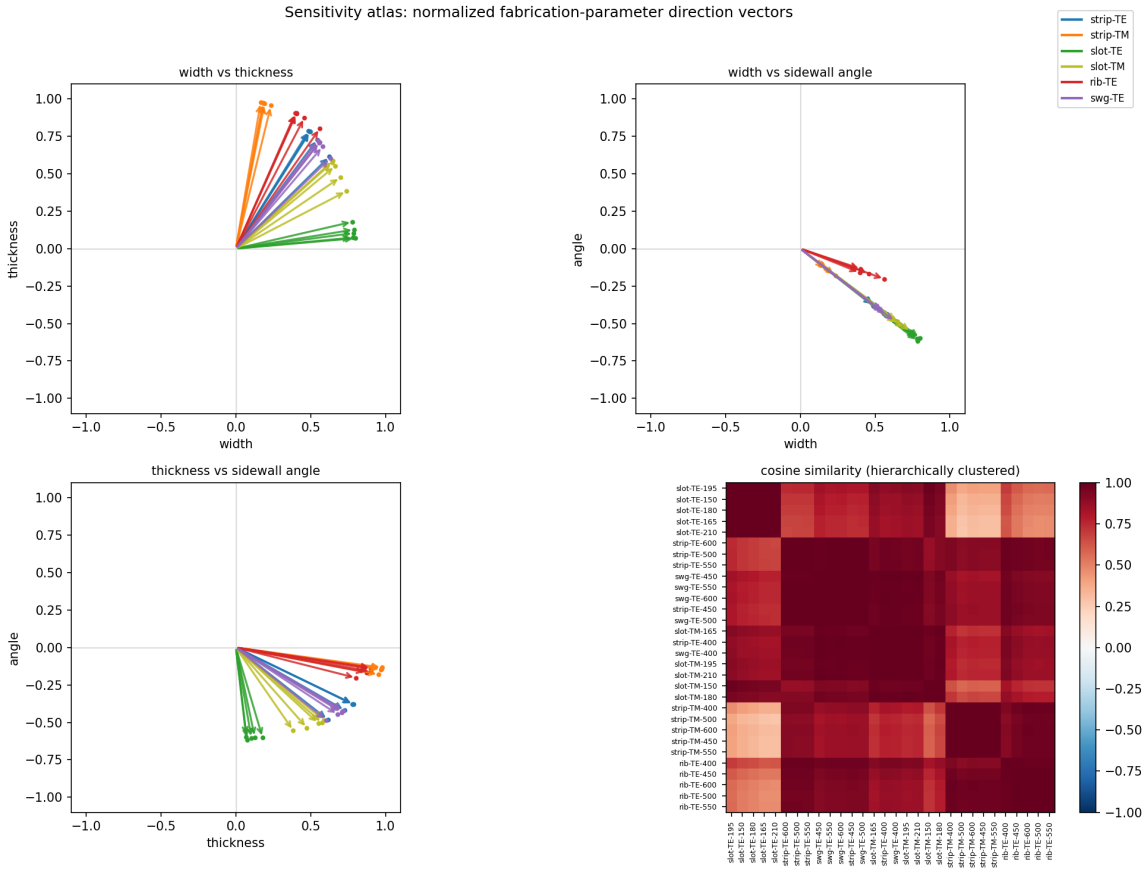


Figure 1. Sensitivity directions for all 30 structures, shown as three two-parameter projections, with a clustered cosine-similarity map. Made with the original averaging code (see the last section).

Existing structures sit in a small corner of the space

A principal component analysis of the direction vectors showed they collapse onto an almost flat two-dimensional sheet (Figure 2). For the first 15 structures (strip and slot), the first two components held 95.3% and 4.7% of the variance, and the third only 0.009%. Adding the rib later raised the third component to about 0.1%, which is the rib moving toward the gap without closing it. That third direction, mostly a mix of width and sidewall angle (loadings -0.60, 0.03, -0.80), was essentially empty. No strip, TM or slot structure reached it. I checked that this wasn't caused by how I scaled the three parameters: across seven different scalings, including none at all, the third component stayed below 0.01%.

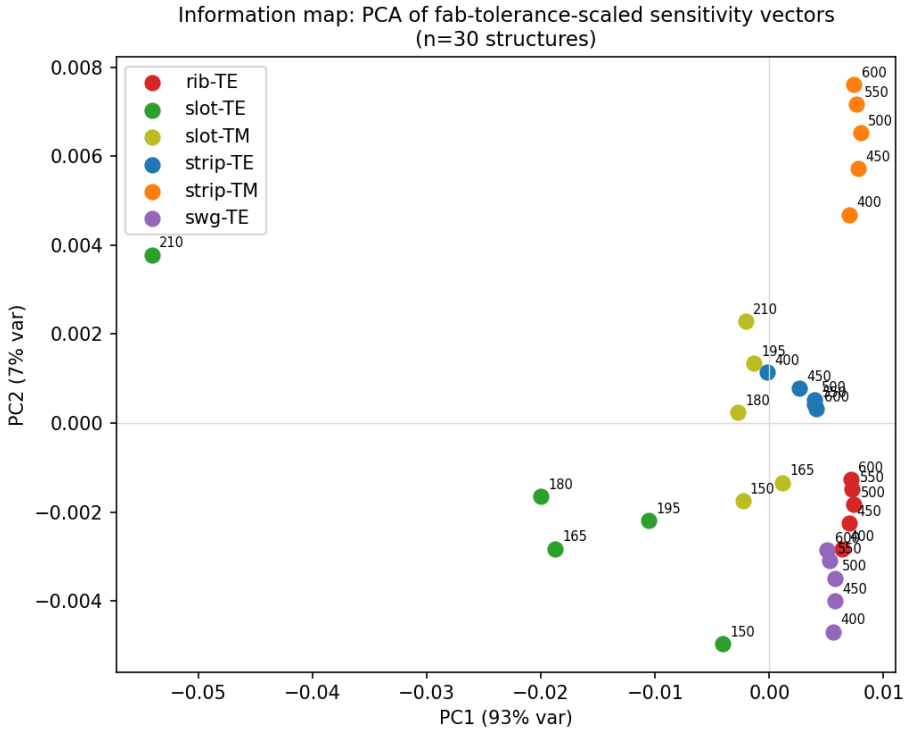


Figure 2. Principal component map of the sensitivity directions, colored by structure type and labeled by width (all 30 structures). Made with the original averaging code.

Slot waveguides helped, but less than they first appeared to

Adding slot waveguides to a strip-only set improved the smallest eigenvalue a lot when I assumed slot rings measure as precisely as strip rings. Slot waveguides are known to be lossier, and two same-process comparisons put their loss at about 3.5 to 5 times that of strip (Baehr-Jones et al., Opt. Express 19, 11529 and 19, 26275, 2011). With a 4x noise penalty the gain dropped to about 7x. A slot TM mode turned out to point in nearly the same direction as an ordinary strip TE mode (about 2° apart), so it added very little.

Slot noise compared to strip	Improvement in smallest eigenvalue
Same (unsourced)	90.8x
4x worse (sourced)	7.1x
10x worse	2.1x

Rib waveguides were the strongest result

A rib (a partially etched waveguide sitting on a thin silicon slab, 90 nm in my model) pointed much further into the empty direction than anything else I tried. Its projection onto that direction was 0.087 to 0.154, against 0.018 to 0.031 for strip, slot and the subwavelength grating. Adding five rib widths to the strip set improved the smallest eigenvalue about 570x. Rib derivatives were actually

smaller in magnitude than strip ones, so the gain came from pointing in a new direction, not from simply being more sensitive. The subwavelength grating, modeled as a uniform effective medium, added nothing (1.1x) and served as a useful negative control.

I checked the rib solver further before trusting this. My own effective-index calculation gave 2.6033 for the 500 nm rib, against 2.602 from a published calculation of the same geometry, and the full solver sits 1.2 to 2.4% below the effective-index value, the same pattern I see for strip. The mesh was also converged to within 0.64% down to 200 nm wide ribs. For noise, one same-study comparison found rib loss about 0.2 times strip loss, though for a wider rib than mine, so I bracketed rib noise between 0.2x and 1.0x of strip. That gives an improvement of 571.7x (conservative) to about 4000x (optimistic), so unlike slot, the rib result did not shrink under realistic noise.

The sidewall-angle precision itself

The real question is about sidewall angle, not the blended worst case, so I also computed the Cramér-Rao bound on angle alone, after accounting for width and thickness being unknown too.

Structure set	Angle precision (1 sigma)	Improvement
Strip baseline (5 widths, ring + MZI)	0.0210°	
+ rib, conservative noise	0.0009°	23x
+ rib, optimistic noise	0.0004°	51x
+ slot, sourced noise	0.0081°	2.6x

With only two strip rings the angle precision was 0.188°. Interestingly, the worst-determined direction in that case was mostly width (component -0.92), not angle (-0.37), which explains why the condition number looked so bad even though angle by itself was not hopeless. Adding a single rib to those two rings improved angle precision 11 to 14x, against 1.9x for a slot and 2.0x for a third strip ring, so the rib advantage held even with very few structures.

Slab thickness matters

The rib benefit depended on the slab thickness. It never went away, but it meant the rib should be treated as a family of designs rather than one fixed recipe.

Slab thickness	Improvement in smallest eigenvalue
50 nm	789x
70 nm	961x
90 nm	834x
110 nm	585x
130 nm	352x
150 nm	182x

All of these numbers come from a linearized bound with assumed noise levels. They describe the best case, not what a real fitting procedure would get. Even strip alone gives sub-0.1° angle precision on paper, which a real extraction clearly does not achieve, so the relative ranking (rib well ahead of slot for sidewall angle) was the part I trusted most.

The follow-up check, and why it was stopped

The rib result had one obvious weak point. The slab thickness was fixed at 90 nm and treated as perfectly known. In a real process the partial etch depth varies from wafer to wafer, and the partial etch may not leave the same sidewall angle as the full etch. If those extra unknowns soak up the information the rib was providing, the rib advantage could be much smaller than it looked. I set up a follow-up study to add etch depth (and optionally a separate sidewall angle for the partial etch) as unknowns and redo the analysis. The new code is in `src/etch_depth_*.py` and its outputs are in `results_etch_depth/`.

While setting this up I also noticed that the code's nominal sidewall is vertical (90°), not 85°, and that the ring noise levels are placeholders without a source. The effective-index noise of 6.5e-6 matches a 10 pm resonance precision only if the group index is ignored; including it gives about 2.6e-5, so that value may be about 4x optimistic.

Before running the Fisher analysis I checked that the new derivatives were stable, by computing each one with three step sizes (h/2, h and 2h) and requiring them to agree within 2%. They didn't. Only 34 of 240 entries passed, and the etch-depth derivatives disagreed by 36 to 41% at every rib.

I traced this to how the existing code builds the material map on the mesh. Where a grid cell straddles a silicon/oxide boundary, the code averages the material over a window equal to the average grid spacing of the whole grid (about 43 x 37 nm), not the actual size of that cell (about 27 x 20 nm near the waveguide). As a result, moving a boundary by 1 nm doesn't add 1 nm worth of silicon to the model. Depending on where the boundary sits relative to the grid, it adds anywhere from about 0.5 to 1.25 nm. All of the slab thicknesses I used happen to sit at spots where the response is about half what it should be. When I rebuilt one rib using each cell's true size, silicon was conserved exactly and the derivatives agreed across step sizes to within 1% (Figure 3).

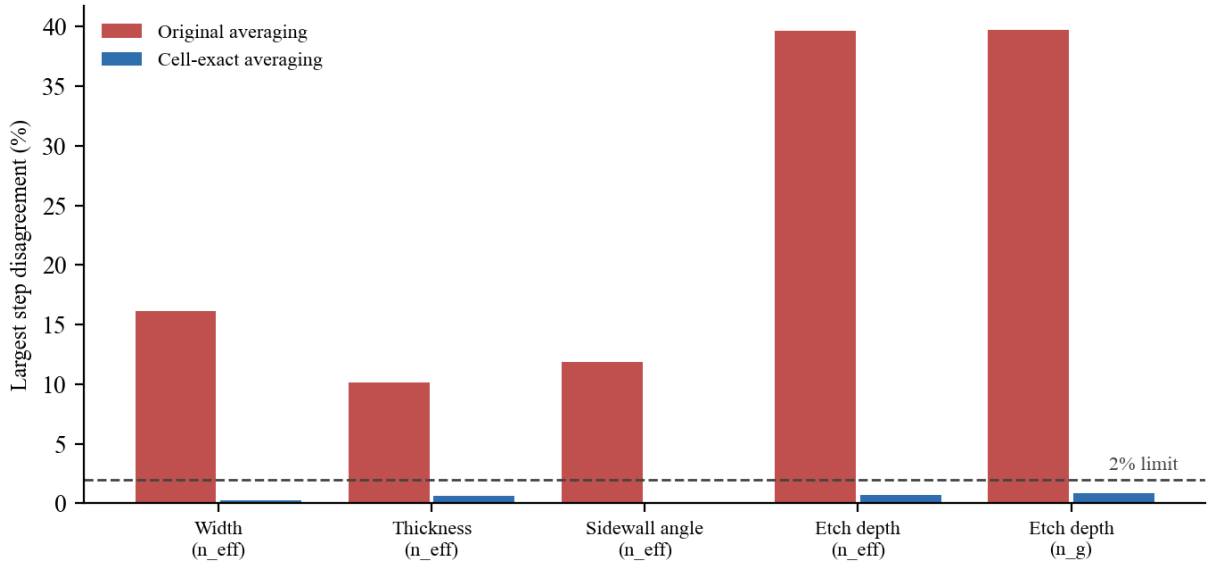


Figure 3. Step-size check for the 500 nm rib on a 90 nm slab: the largest disagreement between steps h/2, h and 2h for each derivative, with the original averaging and with each cell's true size. The dashed line is the 2% limit.

This affects more than the new study. With the averaging fixed, the rib derivatives change by different amounts:

Derivative of effective index (500 nm rib)	Original	Corrected	Change
Sidewall angle	-9.34e-4	-1.524e-3	+63%
Etch depth	-1.07e-3	-1.640e-3	+53%
Width	5.59e-4	8.29e-4	+48%
Thickness	4.20e-3	4.45e-3	+6%

Because each parameter is off by a different amount, the direction of each structure's sensitivity changes, and the rib result was specifically about direction. The strip derivatives go through the same code. So the earlier numbers above, including the rib gains and the empty direction in Figure 2, need to be recomputed with corrected averaging before they can be relied on. One small group-index derivative (with respect to thickness) was still unstable after the fix, because its value is close to zero, and needs a separate check.

I stopped there rather than run the Fisher analysis on numbers I already knew were off. Before continuing, three choices are needed: whether to fix the averaging in the existing code or build corrected versions alongside it, how to judge derivatives that are close to zero, and whether to keep the nominal sidewall at 90° or move to 85°. The detailed numbers are in `results_etch_depth/STOP_REPORT.md`.